



# Balanced and Unbalanced Forces, Introduction



*Physical Science — Forces & Motion | Unit 1, Lesson 1 of 6*



**Grade:** 3rd Grade



**Duration:** 45–60 minutes



**Subject:** Physical Science



*When forces match, motion stays the same. When one force wins, motion changes.*



## NGSS Standard:

**3-PS2-1** — Plan and conduct an investigation to provide evidence of the effects of balanced and unbalanced forces on the motion of an object.



## Learning Objective

Students will investigate balanced and unbalanced forces, make observations during the lesson activity, and explain what they learned using evidence from the lesson.



## The Big Idea

Objects do not just move for no reason. A push or pull changes motion. If forces are balanced, they cancel out and the object does not change motion. If forces are unbalanced, the object's motion changes.



## The Science

A **force** is a push or a pull. Forces can start motion, stop motion, speed something up, slow it down, or change its direction. Scientists compare what happens when forces are equal and when they are unequal.

When two forces are the same size but act in opposite directions, they are **balanced**. Balanced forces do not mean there are no forces. They mean the forces match. When one force is stronger, the forces are **unbalanced**, and the object's motion changes.

## Balanced

Equal pushes or pulls cancel out. The object stays still or keeps moving the same way it already was.

## Unbalanced

One force is stronger. The object starts moving, stops, speeds up, slows down, or changes direction.

### Key vocabulary:

force

push

pull

balanced

unbalanced

motion

evidence



## Materials

- 1 small toy car, block, or smooth object that can slide
- 1 flat table or floor surface
- 2 index cards or stiff paper pieces
- Masking tape
- Pencil
- Printed Student Worksheet
- *Optional bonus:* 1 smooth wrapped hard candy or small bouncy ball (adult supervision for choking hazard around young children)



## Let's Get Started

### Step 1 — Set Up a Fair Test (5 min)

Place the object on the same starting spot each time. Explain that good investigations keep most things the same so we can notice what changed.

### Step 2 — Try a Balanced Push (8 min)

Use two index cards to push gently from opposite sides at the same time. Ask: *"Did the object change motion? What do you notice?"*

#### Simple setup sketch:

[index card] → [ toy car / block ] ← [index card]

*Parent note:* both cards touch the object at the same time and push with the same gentle force.

**Step 3 — Try an Unbalanced Push (8 min)**

Now make one push stronger than the other. Watch what happens. Ask: *"Which force won? How did the motion change?"*

**Step 4 — Repeat and Record (10 min)**

Do at least three trials: one balanced and two unbalanced. Record what happened each time on the Student Worksheet.

**Step 5 — Explain with Evidence (10 min)**

Use your observations to explain why the object stayed still in one test and moved in another. Push students to use the words *balanced*, *unbalanced*, and *evidence*.

**Bonus Wow Experiment — Slippery Squeeze Surprise (5 min)**

Place a **smooth wrapped hard candy** or **small bouncy ball** between two index cards. Squeeze both cards inward gently with equal force and notice that the object may wobble but usually stays near the middle. Then press harder on one side and watch it shoot out toward the weaker side. Ask: *"When did it pop out? What changed about the forces?"*

*Safety note:* Use adult supervision and keep small objects away from younger siblings because they can be a choking hazard.

 **Discussion Questions**

 **Question 1:** What happened when the pushes matched?

*Look for: The forces were balanced, so the object did not change motion.*

 **Question 2:** What happened when one push was stronger?

*Look for: The forces were unbalanced, so the object moved or changed motion.*

 **Question 3:** Why is it important to repeat the test?

*Look for: Repeating gives stronger evidence and helps us see whether the same pattern happens again.*

 **Question 4:** Does balanced mean there were no forces?

*Look for: No. Balanced means the forces were equal and canceled each other out.*

## ✨ Extensions

-  **Sports Spotting:** Find a sports example of balanced forces and one of unbalanced forces.
-  **Change the Object:** Repeat the test with an object that is heavier or rougher. How does that affect your results?
-  **Evidence Sentence:** Write one complete sentence using the words *balanced*, *motion*, and *evidence*.



## Parent/Teacher Notes

- **Keep it visible:** The goal is not precision physics. The goal is helping students notice whether motion changed or stayed the same.
- **Push observation language:** Encourage words like stayed still, moved, changed direction, started moving, and slowed down.
- **Correct a common misconception:** Students often think balanced means “nothing is happening.” Re-emphasize that balanced means the forces match.
- **Assessment check:** A successful response explains both what happened and why the evidence supports balanced or unbalanced forces.



# Student Worksheet — Balanced and Unbalanced Forces



Name: \_\_\_\_\_ Date: \_\_\_\_\_

Part 1 — Record Your Tests

| Test                                                                           | What happened? | Balanced or Unbalanced? |
|--------------------------------------------------------------------------------|----------------|-------------------------|
| Test 1                                                                         |                |                         |
| Test 2                                                                         |                |                         |
| Test 3                                                                         |                |                         |
| Circle the trials that match each other. Which result happened more than once? |                |                         |

Part 2 — Explain

When the forces were balanced, the object \_\_\_\_\_.

When the forces were unbalanced, the object \_\_\_\_\_.

My evidence is \_\_\_\_\_.

Two trials I can compare are \_\_\_\_\_ and \_\_\_\_\_.

Part 3 — Draw One Example

Draw one balanced-force test and one unbalanced-force test.

Draw here 





## Parent Answer Key

**Detach note:** This page is for parent/teacher reference. Student wording may vary if the science idea is correct.

### Part 1

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Answers vary. Balanced trials should show little or no change in motion. Unbalanced trials should show the object moving or changing motion.

### Part 2

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- **Balanced answer:** the object stayed still / did not change motion.
- **Unbalanced answer:** the object moved / changed motion.
- **Strong evidence statement:** the student compares at least two trials and explains what stayed the same or changed.
- **Look for:** observations from the chart, not just the vocabulary words by themselves.

### What Success Looks Like

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- Student can distinguish balanced from unbalanced situations.
- Student uses evidence from the tests rather than only repeating vocabulary.
- Student can explain that balanced does not mean “no forces.”

### Troubleshooting / Misconceptions

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- If the student says balanced means “nothing happened at all,” remind them that equal forces were still acting.
- If a test result looks messy, focus on the pattern across repeated trials rather than one imperfect push.

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# Motion Changes with Force

*Physical Science — Forces & Motion | Unit 1, Lesson 2 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Physical Science



*A stronger push can change how far an object moves. Repeated tests help us spot the pattern.*

## **NGSS Standard:**

**3-PS2-1** — Plan and conduct an investigation to provide evidence of the effects of balanced and unbalanced forces on the motion of an object.

## **Learning Objective**

Students will investigate motion Changes with Force, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

## **The Big Idea**

A force can change an object's motion. When we compare a gentle push to a stronger push, we can collect evidence that stronger and weaker forces can change how far an object moves.

## **The Science**

Scientists look for patterns by keeping most parts of a test the same and changing just one thing. In this lesson, students keep the same object and starting point, but compare different strengths of pushes.

If a stronger push acts on an object, the motion may change more than it does with a weaker push. Repeating the test helps students gather **evidence** instead of guessing from only one try.



## What pattern are we looking for?

When the push gets stronger, the object usually moves farther or faster. The exact numbers may change a little, but the pattern should show up across repeated trials.

### Key vocabulary:

force

motion

distance

stronger force

weaker force

evidence

compare



## Materials

- 1 toy car, small block, or object that slides
- 1 flat table or smooth floor surface
- Masking tape for a starting line
- Ruler or measuring tape
- Pencil
- Printed Student Worksheet



## Let's Get Started

### Step 1 — Plan the Investigation (6 min)

Set up one starting line and choose one object and one surface. Students make a prediction before testing: *Which push strength will move the object farther, and why?*

### Step 2 — Define the Variable (6 min)

Explain that the **strength of the push** is the only thing that should change. Keep the object, starting line, surface, and direction the same so the test stays fair.

### Step 3 — Run 3 Gentle-Push Trials (10 min)

Use a gentle push three times from the same starting line. Measure each distance in inches or centimeters and record all three results in the data table.

**Step 4 — Run 3 Strong-Push Trials (10 min)**

Use a stronger push three times with the same object and setup. Measure and record each distance so students have two sets of comparable data.

**Step 5 — Analyze the Data (8 min)**

Circle the longest distance in each group, then find the average for the gentle pushes and the strong pushes. Students can make a quick bar graph or compare the averages in words.

**Step 6 — Make an Evidence-Based Claim (8 min)**

Write a conclusion using the data: *"My strong pushes usually moved the object farther because..."*  
Students should point to measured distances or averages, not just opinion.

 **Discussion Questions**

 **Question 1:** What pattern do you notice when you compare the average gentle push to the average strong push?

*Look for: The stronger push usually produced a larger average distance.*

 **Question 2:** Why did we keep the object, surface, and starting line the same?

*Look for: Those controls make it a fair test, so push strength is the variable we can compare.*

 **Question 3:** If one trial did not match the pattern, should we ignore all the other data?

*Look for: No. Scientists look at the full set of trials and explain outliers instead of relying on one result.*

 **Question 4:** Which numbers from your table are the best evidence for your claim?

*Look for: Students should cite measured distances, averages, or a clear comparison across several trials.*

 **Extensions**

-  **Average It:** Find the average distance for gentle pushes and for strong pushes.
-  **Change One Variable:** Try a heavier object next. Does the same pattern still appear?
-  **Real Life:** Think of a time when a stronger push or pull changed motion in real life, like opening a door or kicking a ball.



## Parent/Teacher Notes

- **Fair-test reminder:** Keep the object and starting line the same. The lesson works best when students clearly see that push strength is the variable changing.
- **Measurement does not need to be perfect:** The goal is pattern-finding, not exact lab precision.
- **Evidence language matters:** Encourage students to point to measured distances or repeated observations, not just say “because it was stronger.”
- **Assessment check:** Students should be able to make a claim and back it up with data from their chart.



# Student Worksheet — Motion Changes with Force



Name: \_\_\_\_\_ Date: \_\_\_\_\_

Part 1 — Prediction

I predict the \_\_\_\_\_ push will move the object farther because \_\_\_\_\_.

Part 2 — Record Your Tests

| Trial | Type of Push | Distance Moved |
|-------|--------------|----------------|
| 1     | Gentle       |                |
| 2     | Gentle       |                |
| 3     | Gentle       |                |
| 4     | Strong       |                |
| 5     | Strong       |                |
| 6     | Strong       |                |

Average gentle push: \_\_\_\_\_

Average strong push: \_\_\_\_\_

Part 3 — Make a Claim with Evidence

Claim: When the push was stronger, the object \_\_\_\_\_.

Evidence: The average gentle push was \_\_\_\_\_ and the average strong push was \_\_\_\_\_.

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Reasoning: This shows force changed motion because \_\_\_\_\_.

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### Part 4 — Draw or Graph the Data

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Draw one gentle push and one strong push from the same starting line, or sketch a simple two-bar graph for your averages.

Draw here 



## Parent Answer Key

**Detach note:** This page is for parent/teacher reference. Student wording may vary if the science idea is correct.

### Part 1

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Predictions may vary, but students should expect the stronger push to move the object farther.

### Part 2

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Distances vary. Strong pushes should usually move the object farther than gentle pushes across the three-trial set.

Average answers will depend on the family's measurements.

### Part 3

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- **Claim:** when the push was stronger, the object usually moved farther / changed motion more.
- **Evidence to look for:** measured distances, averages, or a comparison across several trials.
- **Reasoning:** a stronger force can change an object's motion more than a weaker force.

### What Success Looks Like

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- Student compares stronger and weaker pushes correctly.
- Student uses evidence from repeated tests or averages instead of guessing.
- Student can explain that push strength was the variable that changed.
- Student understands that keeping the object, surface, and start line the same helps make a fair comparison.

### Troubleshooting / Misconceptions

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- If results are messy, remind students that repeated tests and averages matter more than one outlier.
- If a student says “strong always means faster,” point back to the actual evidence they collected and discuss how we describe what we observed.
- If averaging feels too advanced, compare all six trials verbally and circle the stronger pattern instead.



# Magnets pull without touching

Physical Science — Forces & Motion | Unit 1, Lesson 3 of 6



**Grade:** 3rd Grade



**Duration:** 45–60 minutes



**Subject:** Physical Science



**Kit Lesson**



*Magnets can make some objects move even when they are not touching them.*



## NGSS Standard:

**3-PS2-1** — Plan and conduct an investigation to provide evidence of the effects of balanced and unbalanced forces on the motion of an object.



## Learning Objective

Students will plan a fair magnet investigation, test how distance and material affect magnetic force, and use evidence to explain why some objects move without being touched.



## The Big Idea

Magnetic force can act from a distance, but the strength of the pull changes depending on the material and how far away the magnet is.



## The Science

A magnet creates an invisible magnetic field around it. That field can pull on some materials, especially iron and steel, even before the magnet touches them. The pull is usually stronger when the magnet is closer and weaker when the magnet is farther away or when more material is in the way.

Third grade students can do more than notice that magnets work. They can design a fair test, compare results, and use evidence to answer better questions, such as which materials respond most clearly and how distance changes motion.



**Key vocabulary:**

magnet

magnetic force

magnetic field

distance

evidence

fair test

**Materials**

- 2 bar magnets or refrigerator magnets
- 6–10 test objects such as paperclips, foil, plastic spoon, coin, craft stick, fabric, key, twist tie, eraser
- Try to include a mixed set: one clearly magnetic object (like a paperclip), one metal object that may not be magnetic (like some coins), and one non-metal object.
- 3 index cards or sheets of paper to create different testing barriers/distances
- Ruler
- Pencil
- Printed Student Worksheet

**Let's Get Started****Step 1 — Plan a fair test (5 min)**

Choose 4 to 6 test objects and record a prediction for each one. Decide what will stay the same in every trial: the same magnet, the same starting distance, and the same way of checking for movement.

**Step 2 — Test which materials respond (8 min)**

Bring the magnet close to each object without touching it. Record whether the object moved, did not move, or only moved a little. This gives you stronger evidence than a simple yes or no list.

**Step 3 — Test how distance changes the pull (10 min)**

Use a ruler or stacked index cards to test the same magnetic object from different distances. Try very close, a little farther away, and farther still. Record when the object stops moving. This is the part that levels the investigation up.

**Step 4 — Test through barriers (8 min)**

Place a paperclip under one sheet of paper, then under two or three sheets. Move the magnet above the stack and compare what happens. Ask whether the force still works and whether it feels weaker.

**Step 5 — Make a claim from evidence (10 min)**

Use your notes to write a short claim: magnets can pull some objects without touching, but the effect depends on the material and the distance. Back it up with two pieces of evidence from your investigation.



## Discussion Questions



**Question 1:** Which materials responded most clearly to the magnet?

*Look for: iron or steel objects such as paperclips, some keys, or other strongly magnetic items.*



**Question 2:** What changed when the magnet was farther away?

*Look for: the pull became weaker, the object moved less, or it stopped moving altogether.*



**Question 3:** Why is this a better investigation than testing only one object one time?

*Look for: students gathered more evidence, compared conditions, and looked for patterns instead of making one observation.*



**Question 4:** What claim can you make about magnetic force from your evidence?

*Look for: magnetic force can act without touching, but it depends on the object and distance.*



## Extensions

- **Distance graph:** Make a simple bar graph showing how many paper layers or how much distance the magnet could still work through.
- **Compare two magnets:** Test whether one magnet can pull from farther away than another.
- **Evidence paragraph:** Write a short explanation using the words *magnetic field*, *distance*, and *evidence*.



## Parent/Teacher Notes

- **Safety note:** Keep magnets away from electronics, credit cards, and anything medically sensitive.
- **Choose a mixed test set:** Include one object you expect to be magnetic, one metal object that may not be magnetic, and one clearly non-metal object so the pattern is easier to see.
- **Mixed results are useful:** If a key, coin, or household metal item gives an unexpected result, treat that as good science evidence to talk about rather than a lesson failure.
- **Misconception check:** Students often think all metals are magnetic. Let the evidence show that only some are.
- **Assessment check:** Students should explain that the object's motion is evidence of a force acting from a distance.



# Student Worksheet — Magnets Pull Without Touching



Name: \_\_\_\_\_ Date: \_\_\_\_\_

Part 1 — Predict and Test

| Object              | My Prediction | What Happened? |
|---------------------|---------------|----------------|
| Paperclip           |               |                |
| Coin                |               |                |
| Plastic spoon       |               |                |
| Key or metal object |               |                |

Part 2 — What Is the Pattern?

The objects that were magnetic had this in common: \_\_\_\_\_.

This investigation shows a force can act without touching because \_\_\_\_\_.

Part 3 — Draw It

Draw the paper test or one object moving toward the magnet.

Draw here 



## Parent Answer Key

**Detach note:** This page is for parent/teacher reference. Student wording may vary if the science idea is correct.

### Part 1

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Predictions vary. Paperclips and some steel objects should be attracted. Plastic, wood, fabric, and many coins should not be.

### Part 2

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- Objects that were magnetic usually contained **iron or steel**.
- The investigation shows a force can act without touching because **the object moved before direct contact happened**.

### What Success Looks Like

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- Student can describe magnetism as a force that can act from a distance.
- Student can identify that only some materials are magnetic.
- Student uses observations as evidence.

### Troubleshooting / Misconceptions

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- If a student says all metal is magnetic, compare results from a paperclip and a coin.
- If a key or coin gives a surprising result, remind students that household objects can vary and that the evidence still helps us build the pattern.
- If an object response is weak, move the magnet closer and repeat the test carefully.

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# Magnetic Patterns

*Physical Science — Forces & Motion | Unit 1, Lesson 4 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Physical Science



*When we test many objects, patterns appear. Scientists use those patterns to explain what magnets do.*

## **NGSS Standard:**

**3-PS2-1** — Plan and conduct an investigation to provide evidence of the effects of balanced and unbalanced forces on the motion of an object.

## **Learning Objective**

Students will investigate magnetic Patterns, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

## **The Big Idea**

Scientists look for patterns in results. When we test many objects, we can find a pattern about which materials are magnetic and which are not.

## **The Science**

One test result can be interesting, but many test results can reveal a pattern. Magnets usually attract materials that contain iron or steel. They do not attract most wood, plastic, fabric, paper, or many coins.

Finding a pattern helps scientists make explanations and predictions. If students notice that the same kinds of materials respond to magnets over and over, they can use evidence to explain magnetism more clearly.

**Key vocabulary:** pattern, magnetic, material, evidence, compare, predict



## Materials

- 2 bar magnets
- 8–12 test objects of mixed materials
- Sorting tray or table space
- Pencil
- Printed Student Worksheet



## Let's Get Started

### Step 1 — Review the Last Lesson (5 min)

Briefly remind students that magnets can pull some objects without touching them. Tell them that today they are looking for a pattern in the evidence.

### Step 2 — Test More Objects (10 min)

Test a larger set of objects and record whether each one is magnetic or not magnetic.

### Step 3 — Sort by Material (10 min)

Group your objects by what they seem to be made of: metal, plastic, wood, fabric, paper, or something else. Compare the groups.

### Step 4 — Look for the Pattern (10 min)

Ask: Which kinds of materials were magnetic most often? Which kinds were not?

### Step 5 — Make a Claim (10 min)

Use the pattern in your chart to make a claim about what magnets usually attract.



## Discussion Questions



**Question 1:** What pattern did you notice?

*Look for: certain metal objects, especially iron or steel ones, were magnetic more often than other materials.*



**Question 2:** Why do scientists test many objects instead of just one?

*Look for: many tests give stronger evidence and help reveal a pattern.*

**Question 3:** Could you predict whether a new object might be magnetic?

*Look for: Yes, if you know the material, you can make a better prediction from the pattern.*

**Extensions**

-  **Classroom Sort:** Find five new objects and predict whether they will be magnetic before testing them.
-  **Tricky Metals:** Test different coins or metal tools and talk about why results may differ.
-  **Claim and Evidence:** Write one claim sentence and one evidence sentence about magnetism.

**Parent/Teacher Notes**

- **Focus on patterns, not perfection:** The key is noticing repeated evidence across many objects.
- **Choose a mixed object set:** Include clearly magnetic items, clearly non-magnetic nonmetals, and at least one metal object that may surprise students so the pattern is built from evidence.
- **Mixed results are useful:** If a coin, key, or tool does not act the way a student predicted, use that result to refine the pattern instead of treating it as a mistake.
- **Good misconception target:** Students may think shiny means magnetic. Let the evidence correct that idea.
- **Assessment check:** Students should make a claim about magnetic materials and support it with examples from the chart.



# Student Worksheet — Magnetic Patterns



Name: \_\_\_\_\_ Date: \_\_\_\_\_

Part 1 — Test Objects

| Object   | Material | Magnetic or Not? |
|----------|----------|------------------|
| Object 1 |          |                  |
| Object 2 |          |                  |
| Object 3 |          |                  |
| Object 4 |          |                  |
| Object 5 |          |                  |

Part 2 — Find the Pattern

The pattern I noticed is \_\_\_\_\_.

I predict this new object would be magnetic / not magnetic because \_\_\_\_\_.

Part 3 — Draw or Label

Draw one magnetic object and one non-magnetic object.

Draw here 





## Parent Answer Key

**Detach note:** This page is for parent/teacher reference. Student wording may vary if the science idea is correct.

### Part 1

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Answers vary by objects tested. Magnetic objects should generally be iron- or steel-based items.

### Part 2

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- The pattern may be described as: **many iron or steel objects were magnetic, but most plastic, wood, paper, and fabric objects were not.**
- A good prediction uses the material as evidence.

### What Success Looks Like

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- Student can identify a pattern across multiple tests.
- Student uses evidence to support a prediction.
- Student avoids claiming that all metals are magnetic.

### Troubleshooting / Misconceptions

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- If students overgeneralize, ask them to point to the exact examples in the chart.
- If a household object gives a mixed or surprising result, use that as evidence for refining the pattern rather than ignoring it.
- If materials are unclear, let students write their best guess and discuss how scientists revise ideas with more evidence.

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# Design a Magnetic Game or Tool

Physical Science — Forces & Motion | Unit 1, Lesson 5 of 6



**Grade:** 3rd Grade



**Duration:** 50–65 minutes



**Subject:** Physical Science



**Kit Lesson**



*Engineers can use magnetic force to solve problems or make fun games.*



## **NGSS Standard:**

**3-PS2-1** — Plan and conduct an investigation to provide evidence of the effects of balanced and unbalanced forces on the motion of an object.



## **Learning Objective**

Students will plan or test ideas for design a Magnetic Game or Tool, make observations during the activity, and explain their choices using evidence from the lesson.



## **The Big Idea**

We can use what we know about magnetic force to design something that moves objects without touching them directly.



## **The Science**

Engineers use scientific ideas to solve problems. Since magnets can pull some objects from a short distance, students can use magnetic force to make a game, picker-upper tool, or simple challenge device.

A strong design should have a clear goal and use materials on purpose. Students should plan, build, and test whether the magnetic force works the way they hoped.

**Key vocabulary:** design, criteria, constraint, magnetic force, test, improve, prototype



## Materials

- 2 bar magnets or refrigerator magnets
- Paperclips, washers, or other magnetic pieces
- String, tape, cardboard, cups, index cards
- Scissors
- Pencil
- Printed Student Worksheet



## Let's Get Started

### Step 1 — Choose a Goal (5 min)

Decide whether you will build a magnetic game or a magnetic tool. Your design should solve one clear problem or create one clear challenge.

### Step 2 — Plan Your Prototype (10 min)

Sketch what you will build and label the parts. Decide where the magnet will go and what magnetic objects it will move. Also name one success criterion and one constraint before you start building.

### Step 3 — Build It (15 min)

Use your materials to make the first version of your design.

### Step 4 — Test It (10 min)

Try the design at least two times. Does it work the way you planned?

### Step 5 — Record What Happened (10 min)

Write what worked, what did not work, and one idea for improving the next version.



## Discussion Questions



**Question 1:** What job is your design supposed to do?

*Look for: a clear purpose, like picking up objects or moving pieces in a game.*



**Question 2:** How does the magnet help your design work?

*Look for: the magnet pulls the magnetic object without direct contact.*

**Question 3:** What part of your first design needs improvement?

*Look for: an honest observation about what failed or worked weakly.*

**Extensions**

-  **Challenge Mode:** Add rules or scoring to your magnetic game.
-  **Real Tool:** Redesign your prototype as a tool that helps pick up small metal objects.
-  **Make It Better:** Add labels, decoration, or clearer instructions for another player.

**Parent/Teacher Notes**

- **Focus on design thinking:** Students do not need a perfect final product on the first try.
- **Criteria matter:** Keep asking whether the design actually solves the chosen problem.
- **Add a simple constraint:** Ask students to stay within the supplied materials or a small build space so the challenge stays concrete.
- **Planning should be specific enough to test:** A strong plan names what the tool/game should do, what counts as success, and one limit such as time, materials, or size.
- **Assessment check:** Students should explain how magnetic force is part of the design.

# Student Worksheet — Design a Magnetic Game or Tool

Name: \_\_\_\_\_ Date: \_\_\_\_\_

## Part 1 — My Goal

I am designing a magnetic game / tool that will \_\_\_\_\_.

My success criterion is \_\_\_\_\_.

My constraint is \_\_\_\_\_.

## Part 2 — My Plan

Draw and label your design.

Draw here 

Parts I must include: \_\_\_\_\_

## Part 3 — Test It

What worked? \_\_\_\_\_

What needs improvement? \_\_\_\_\_

My next change will be \_\_\_\_\_





## Parent Answer Key

**Detach note:** This page is for parent/teacher reference. Student wording and designs may vary.

### Acceptable Responses

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- Goal should be clear and connected to magnetic force.
- Success criterion should explain what counts as working.
- Constraint can name a limit such as materials, size, or time.
- Plan should show where the magnet and magnetic pieces are used.
- Test notes should identify one success and one needed improvement.

**Quick exemplar:** “My tool will pick up two paperclips without touching them. My success criterion is that it works two times in a row. My constraint is that I can only use one magnet, string, tape, and cardboard.”

### What Success Looks Like

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- Student connects the design to a real purpose or game goal.
- Student can explain how the magnet moves or picks up the object.
- Student identifies at least one improvement for the next version.

### Troubleshooting / Misconceptions

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- If the design does not work, treat that as useful test information, not failure.
- If a student uses non-magnetic pieces, redirect them to choose magnetic materials.

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# Test and Improve Your Force Design

*Physical Science — Forces & Motion | Unit 1, Lesson 6 of 6*

 **Grade:** 3rd Grade

 **Duration:** 50–65 minutes

 **Subject:** Physical Science



*Engineers test, learn from evidence, and improve their designs.*

## **NGSS Standard:**

**3-PS2-1** — Plan and conduct an investigation to provide evidence of the effects of balanced and unbalanced forces on the motion of an object.

## **Learning Objective**

Students will test and improve your Force Design, compare what changed, and explain what the results show.

## **The Big Idea**

Designing is not just building once. Engineers test, gather evidence, and improve their designs so they work better.

## **The Science**

A good test checks whether a design meets its goal. Students can compare version 1 and version 2 of a magnetic game or tool by asking what changed and whether the new version works better.

Evidence from tests should guide the redesign. This helps students connect scientific investigation to engineering improvement.

**Key vocabulary:** test, improve, redesign, evidence, criteria, compare





## Materials

- Your magnetic game or tool from Lesson 5
- Magnets and magnetic pieces
- Tape, cardboard, string, scissors, extra supplies for redesign
- Pencil
- Printed Student Worksheet



## Let's Get Started

### Step 1 — Review the Goal (5 min)

Look back at your original design goal. What was your design supposed to do?

### Step 2 — Test Version 1 (10 min)

Run your design at least two times and record what happened.

### Step 3 — Choose One Improvement (10 min)

Use your evidence to decide on one or two changes that should help the design work better.

### Step 4 — Build Version 2 (15 min)

Make the improvement and label your new version.

### Step 5 — Compare Versions (10 min)

Test version 2. Did the design improve? Use evidence to explain your answer.



## Discussion Questions



**Question 1:** What problem did your first design have?

*Look for: a specific issue, such as weak magnet pull, pieces falling, or unclear rules.*



**Question 2:** What evidence helped you choose your redesign?

*Look for: observations from testing, not just guesses.*



**Question 3:** Did version 2 work better? How do you know?

*Look for: a comparison between the first and second versions using evidence.*

## Extensions

-  **Share It:** Explain your redesign to a family member or classmate.
-  **New Goal:** Add a harder challenge or new rule to your design.
-  **Engineer Reflection:** Write one sentence about how evidence helps engineers improve ideas.

## Parent/Teacher Notes

- **Normalize redesign:** Improvement is a success, not a sign the first attempt failed.
- **Push evidence language:** Ask what exactly happened during testing that led to the redesign.
- **Assessment check:** Students should compare versions and support the comparison with evidence.



# Student Worksheet — Test and Improve Your Force Design

Name: \_\_\_\_\_ Date: \_\_\_\_\_

## Part 1 — Test Version 1

What happened when you tested your first design? \_\_\_\_\_

What problem did you notice? \_\_\_\_\_

Evidence from version 1: \_\_\_\_\_

## Part 2 — My Improvement

The change I made was \_\_\_\_\_

I made this change because \_\_\_\_\_

## Part 3 — Compare Versions

Version 1 did \_\_\_\_\_, but version 2 did \_\_\_\_\_.

Version 2 worked better / the same / worse because \_\_\_\_\_.

My proof is \_\_\_\_\_.

Draw version 2 or label the improved part.

Draw here 



## Parent Answer Key

**Detach note:** This page is for parent/teacher reference. Student wording may vary.

### Acceptable Responses

---

- Students should identify a clear problem from testing.
- The improvement should connect to that problem.
- The comparison should use evidence from version 1 and version 2.
- Strong responses name what version 1 did, what changed in version 2, and what proof shows whether version 2 was better, worse, or the same.

### What Success Looks Like

---

- Student uses evidence to justify the redesign.
- Student can compare two versions clearly.
- Student sees redesign as part of engineering.

### Troubleshooting / Misconceptions

---

- If version 2 still does not work well, that is still useful evidence.
- If students choose random changes, ask them to connect the change to a specific problem from testing.

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# What Is a Life Cycle?

Life Science — Life Cycles — Unit 1, Lesson 1 of 6

 **Grade:** 3rd Grade

 **Duration:** 45–55 minutes

 **Subject:** Life Science — Life Cycles

## ONE BIG PATTERN

**Life cycles can look very different, but they all show the same science idea.** A living thing begins life, grows and changes, becomes an adult that can reproduce, and eventually dies.

Birth



Growth



Reproduction



Death

Butterflies, frogs, bean plants, and humans all fit this pattern, even though their stage names and changes are different.

## NGSS Standard:

**3-LS1-1** — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

## Learning Objective

Students will *compare* several organisms, identify the big pattern all life cycles share, and *build a model* that uses one organism's stages as evidence for birth, growth, reproduction, and death.

## The Big Idea



Every living thing has a **life cycle**, but not every life cycle looks the same. A frog, a butterfly, a bean plant, and a human all move through different stages. Even so, they all share the same big idea: **life cycles have a pattern.**

**This is a preview. Unlock the full lesson.**

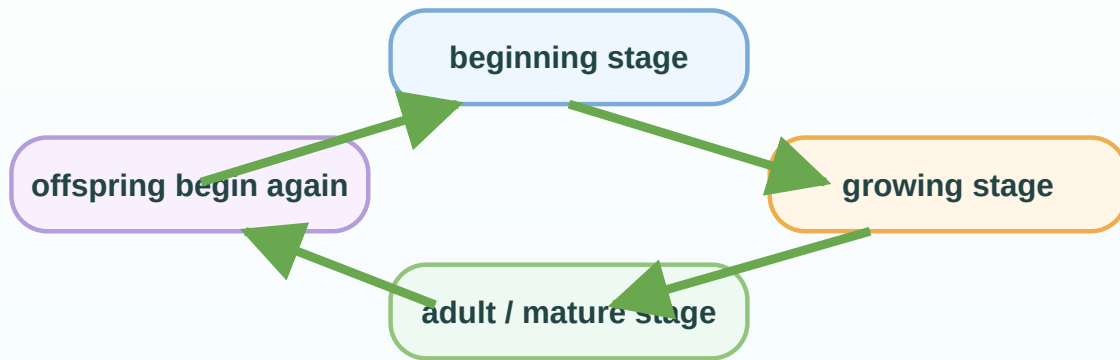
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## SCIENCE SNAPSHOT



Life cycles vary in detail, but they all track how living things grow and make more of their kind.

### What is happening?

A life cycle is the sequence of stages an organism goes through as it grows, survives, and reproduces. Different organisms look different at each stage, but the big pattern is that living things begin, grow, and eventually make new offspring.

### Why it works that way

Life cycles matter because they help scientists compare organisms and understand what each stage is for. Some stages are specialized for feeding, moving, dispersing, or reproducing, so the shape and behavior of the organism may change over time.

### Look for this in the lesson

Students should focus on the repeating pattern, not just memorizing stage names. The science becomes deeper when they explain what changes and what stays part of the same cycle.

**Ask while you read:** How does this organism change across stages, and what big pattern stays the same?



## Compare These Life Cycles

Read each row from left to right. Notice what changes, and notice what stays the same about the big pattern.

### Butterfly

 Butterfly egg on a leaf.

 Butterfly caterpillar on a leaf.



 Butterfly chrysalis hanging from a twig.

 Adult butterfly on a flower.

**This is a preview. Unlock the full lesson.**

You're seeing a preview. Full-library access is not live yet. For now, browse the live store products and free sample lessons while the larger subscription offer stays in Coming Soon.

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Tag

Category

Class

Adult butterfly

This cycle shows metamorphosis, the process of changing from one form to another.

[See Live Products & Free Samples](#)

## Frog

 Frog eggs in water.

Eggs



 Tadpole swimming in water.

Tadpole



 Froglet with legs and a tail.

Froglet



 Adult frog on a lily pad.

Adult frog

It begins in water. As the frog grows, legs develop and the tail becomes smaller.

## Bean Plant

 Bean seed beginning to sprout.

Seed



 Bean seedling with first leaves.

Seedling



 Young bean plant growing taller.

Young plant

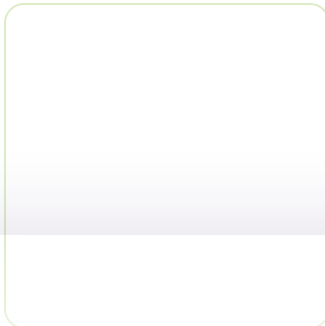


 Mature bean plant with flowers and bean pods.

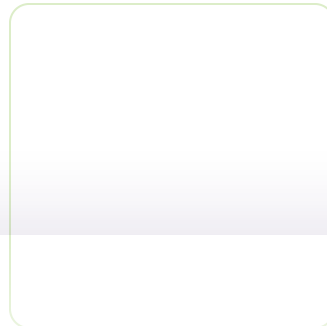
Mature plant

Plants have life cycles too. The mature plant makes flowers and seeds that start the pattern again.

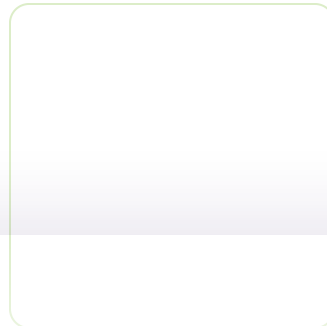
## Human



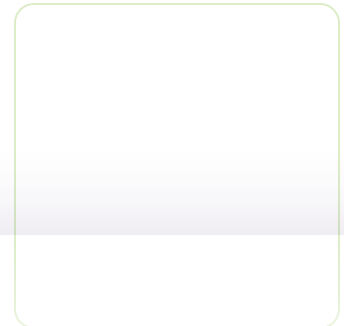
Beginning of life



Growth



Can reproduce



A new life begins

The adult does not turn back into a baby. The cycle continues when adults produce and a new baby begins life.



**This is a preview. Unlock the full lesson.**

### Look for These Two Big Ideas

- Some organisms have life cycles that repeat over and over again.
- All organisms still follow the same broad pattern of growth and death.
- Adults produce offspring that start the cycle in the first stage.

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# Tracking Change Through One Life Cycle

*Life Science — Life Cycles | Unit 1, Lesson 2 of 6*

 **Grade:** Grade 3

 **Duration:** 45–60 minutes

 **Subject:** Life Science

## FROM BROAD PATTERN TO EVIDENCE

In Lesson 1, students learned that all life cycles share the same big pattern: **birth, growth, reproduction, and death**. In this lesson, they track one organism across time and decide when the organism is still the same individual changing and when a *new generation* begins.

*The key idea: growth changes one organism over time, but reproduction starts a new organism and keeps the cycle going.*

## NGSS Standard:

**3-LS1-1** — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

## The Big Idea

One organism can change a lot as it grows, but it is still the same organism. A life cycle becomes a *cycle* when an adult reproduces and a new young organism begins the pattern again.

## The Science



# Plant Life Cycles and Seeds

*Life Science — Life Cycles | Unit 1, Lesson 3 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Life Science — Life Cycles



*Seeds, seedlings, flowers, and new seeds show a plant life cycle in motion.*

## NGSS Standard:

**3-LS1-1** — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

## Learning Objective

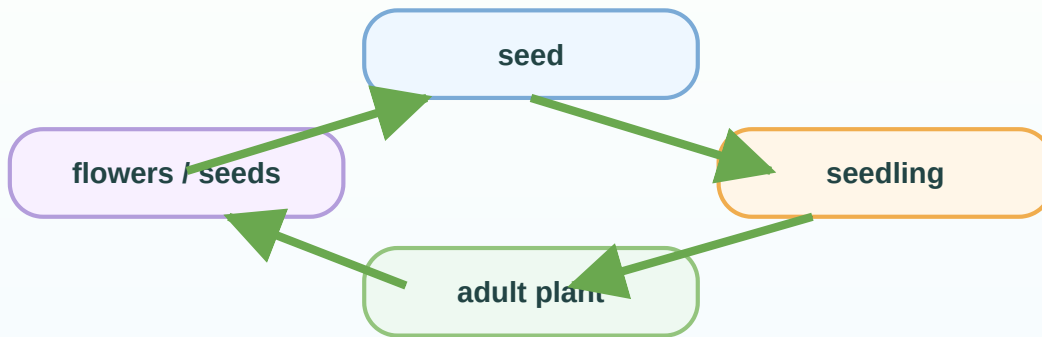
Students will model a plant life cycle and explain where birth, growth, reproduction, and death appear in that model.

## The Big Idea

Plants may look different from animals, but they still follow the same big pattern of birth, growth, reproduction, and death. A seed begins life, a seedling grows, a mature plant reproduces by making new seeds, and the cycle can begin again.



## SCIENCE SNAPSHOT



A plant life cycle shows how a seed can become a plant that produces new seeds.

### What is happening?

Plant life cycles include stages such as seed, seedling, adult plant, flower, fruit, and new seeds. The exact timing varies, but the major scientific idea is that plants make offspring through seeds or other reproductive structures.

### Why it works that way

Seeds protect the embryo and often store food for early growth. Flowers and fruits help with pollination, protection, and dispersal, which increases the chance that the next generation can grow successfully.

### Look for this in the lesson

Students should connect each plant stage to a job, such as germination, growth, flowering, or seed making. That helps the cycle feel meaningful rather than like a list to memorize.

**Ask while you read:** Which stage in the plant cycle helps begin the next generation, and how?



# Animal Life Cycles and Metamorphosis

*Life Science — Life Cycles | Unit 1, Lesson 4 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Life Science — Life Cycles



*Some animals change gradually, and some change dramatically through metamorphosis.*

## NGSS Standard:

**3-LS1-1** — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

## Learning Objective

Students will compare animal life cycles and explain how metamorphosis and gradual growth can both fit the same shared life-cycle pattern.

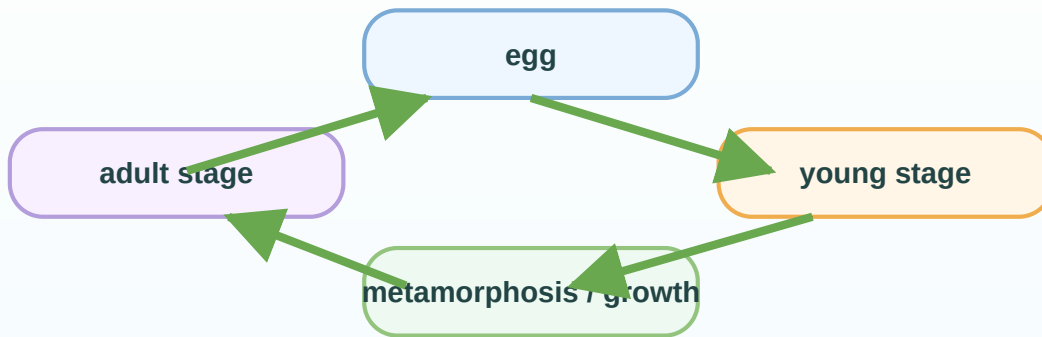


## The Big Idea

Animal life cycles can look very different from one another. Some animals hatch from eggs, some are born live, and some go through metamorphosis, which means their bodies change a lot as they grow. Even with those big differences, animal life cycles still follow the same big pattern.



## SCIENCE SNAPSHOT



Metamorphosis shows that life-cycle stages can have very different bodies and jobs.

### What is happening?

Animal life cycles can include gradual growth or dramatic metamorphosis. In metamorphosis, the body changes substantially between stages because different stages can solve different survival problems.

### Why it works that way

A caterpillar is specialized for eating and growing, while an adult butterfly is specialized for flight and reproduction. Tadpoles and frogs also show how structure and habitat can change across a life cycle.

### Look for this in the lesson

Students should notice that a dramatic change does not mean the organism became something unrelated. It is still the same organism moving through its life cycle.

**Ask while you read:** Why might it help an organism to have different body forms at different life stages?



# Comparing Life Cycle Patterns

*Life Science — Life Cycles | Unit 1, Lesson 5 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Life Science — Life Cycles



*Scientists compare patterns to explain what is shared and what is unique.*

 **NGSS Standard:**

**3-LS1-1** — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

## Learning Objective

Students will compare plant and animal life-cycle models and use evidence to explain what stays the same and what changes across organisms.

## The Big Idea

Scientists do not just memorize stage names. They compare models to look for patterns and explain with evidence. In this lesson, students compare plant and animal life cycles more directly so they can explain what stays the same across very different organisms.



## SCIENCE SNAPSHOT

### Shared life-cycle pattern

- begin
- grow
- reproduce

compare

— — — — —

### Different details

- stage names
- body changes
- timing / habitat

Comparisons help students see the shared pattern behind different kinds of life cycles.

### What is happening?

Comparing life cycles helps scientists see both similarities and differences across organisms. Plants and animals may have different stages, but both show growth, development, and reproduction.

### Why it works that way

A comparison is strongest when students look for patterns in function, not only appearance. One stage may focus on growth, another on dispersal, and another on reproduction even if the organism looks very different from another species.

### Look for this in the lesson

Students should practice saying what is the same at a big-pattern level and what is different in the stage details. That is what makes comparison more scientific than just noticing surface features.

**Ask while you read:** What is one important similarity and one meaningful difference between the cycles you compared?



# Model and Explain a Life Cycle

*Life Science — Life Cycles | Unit 1, Lesson 6 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Life Science — Life Cycles



*A final model should show stage order and explain the shared pattern clearly.*

## NGSS Standard:

**3-LS1-1** — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

## Learning Objective

Students will create a life-cycle model for one organism and use that model as evidence to explain birth, growth, reproduction, and death.

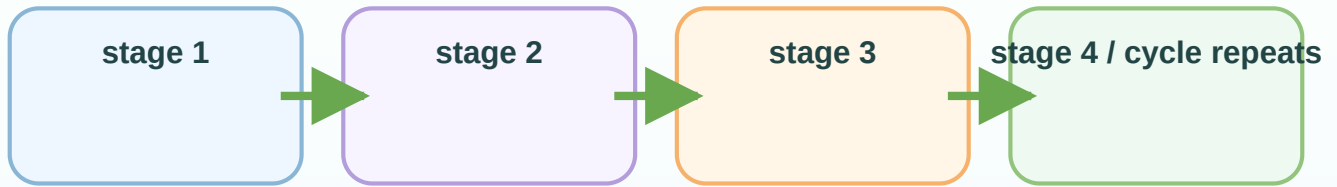
## The Big Idea

A scientific model is not just a neat drawing. It is a tool for showing relationships and explaining ideas. In this culminating lesson, students create and explain a life-cycle model that shows the shared pattern and uses evidence from the unit.





## SCIENCE SNAPSHOT



A strong model helps other people understand how the organism's stages connect in time and purpose.

### What is happening?

A life-cycle model should show the sequence of stages and explain how each stage contributes to survival or reproduction. Good models simplify reality, but they still keep the key pattern clear.

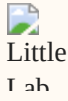
### Why it works that way

Scientists use models to communicate ideas that unfold over time. Arrows, labels, and notes about what happens at each stage make the model much more useful than a picture with no explanation.

### Look for this in the lesson

Students should treat the model as an explanation tool. The model is strongest when it shows both order and function.

**Ask while you read:** Does your model explain what happens at each stage or only show what it looks like?



Little Lab Coats

# Inherited Traits All Around Us

*Life Science — Traits & Inheritance | Unit 2, Lesson 1 of 6*



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits & Inheritance



*Parents and offspring often share visible traits.*



**NGSS Standard:**

**3-LS3-1** — Analyze and interpret data to provide evidence that plants and animals have traits inherited from parents and that variation of these traits exists in a group of similar organisms.

## GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

Lesson 5

Lesson 6



## Learning Objective

Students will identify observable traits and explain that many traits are inherited from parents.



Little Lab Coats

# Plant Traits From Parent Plants

Life Science — Traits & Inheritance | Unit 2, Lesson 2 of 6



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits & Inheritance



*Parent plants and offspring plants often share key traits.*



**NGSS Standard:**

**3-LS3-1** — Analyze and interpret data to provide evidence that plants and animals have traits inherited from parents and that variation of these traits exists in a group of similar organisms.

## GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

Lesson 5

Lesson 6



## Learning Objective

Students will compare offspring plants with parent plants and describe inherited plant traits.



Little Lab Coats

# Animal Traits From Parent Animals

Life Science — Traits & Inheritance | Unit 2, Lesson 3 of 6



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits & Inheritance



*Animal families often show inherited body traits.*



**NGSS Standard:**

**3-LS3-1** — Analyze and interpret data to provide evidence that plants and animals have traits inherited from parents and that variation of these traits exists in a group of similar organisms.

## GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

**Lesson 3**

Lesson 4

Lesson 5

Lesson 6



## Learning Objective

Students will analyze simple animal data to identify inherited animal traits.



## Variation Within a Similar Group

Life Science — Traits & Inheritance | Unit 2, Lesson 4 of 6



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits & Inheritance



*A group can share traits and still show variation.*



**NGSS Standard:**

**3-LS3-1** — Analyze and interpret data to provide evidence that plants and animals have traits inherited from parents and that variation of these traits exists in a group of similar organisms.

### GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

**Lesson 4**

Lesson 5

Lesson 6



### Learning Objective

Students will identify variation in traits within a group of similar organisms.



Little Lab Coats

## Trait Data Detective

*Life Science — Traits & Inheritance | Unit 2, Lesson 5 of 6*



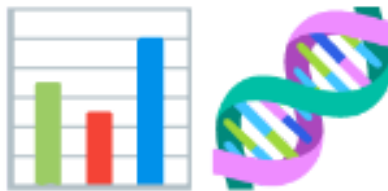
**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits & Inheritance



*Data helps scientists explain trait patterns clearly.*



**NGSS Standard:**

**3-LS3-1** — Analyze and interpret data to provide evidence that plants and animals have traits inherited from parents and that variation of these traits exists in a group of similar organisms.

### GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

Lesson 5

Lesson 6



### Learning Objective

Students will interpret a simple class data set to support claims about inherited traits and variation.



Little Lab Coats

## Make the Inheritance Case

*Life Science — Traits & Inheritance | Unit 2, Lesson 6 of 6*



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits & Inheritance



*Scientists explain inheritance and variation with evidence.*



**NGSS Standard:**

**3-LS3-1** — Analyze and interpret data to provide evidence that plants and animals have traits inherited from parents and that variation of these traits exists in a group of similar organisms.

### GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

Lesson 5

**Lesson 6**



### Learning Objective

Students will analyze and interpret trait evidence to explain inheritance and variation in a final model or argument.



## Variation Can Help Survival

*Life Science — Traits for Survival | Unit 3, Lesson 1 of 6*



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits for Survival



*In the right environment, a variation can become an advantage.*



**NGSS Standard:**

**3-LS4-2** — Use evidence to construct an explanation for how the variations in characteristics among individuals of the same species may provide advantages in surviving, finding mates, and reproducing.



### Learning Objective

Students will explain that variation in traits can give some individuals a survival edge in a specific environment.





# Camouflage and Staying Hidden

Life Science — Traits for Survival | Unit 3, Lesson 2 of 6



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits for Survival



*Camouflage works best when the trait matches the background.*



**NGSS Standard:**

**3-LS4-2** — Use evidence to construct an explanation for how the variations in characteristics among individuals of the same species may provide advantages in surviving, finding mates, and reproducing.

## GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

Lesson 5

Lesson 6



## Learning Objective

Students will use evidence to explain how camouflage trait differences can help organisms avoid predators or capture prey.



Little Lab Coats

# Beaks, Mouths, and Getting Food

Life Science — Traits for Survival | Unit 3, Lesson 3 of 6



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits for Survival



*Different feeding traits work best with different foods.*



**NGSS Standard:**

**3-LS4-2** — Use evidence to construct an explanation for how the variations in characteristics among individuals of the same species may provide advantages in surviving, finding mates, and reproducing.

## GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

**Lesson 3**

Lesson 4

Lesson 5

Lesson 6



## Learning Objective

Students will compare feeding traits and explain how certain variations may help individuals get food in different settings.



## Movement Traits in Different Places

*Life Science — Traits for Survival | Unit 3, Lesson 4 of 6*



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits for Survival



*Movement traits become advantages when they match the environment.*



**NGSS Standard:**

**3-LS4-2** — Use evidence to construct an explanation for how the variations in characteristics among individuals of the same species may provide advantages in surviving, finding mates, and reproducing.

### GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

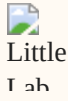
Lesson 5

Lesson 6



### Learning Objective

Students will explain how trait differences in movement can help organisms in different environments.



Little Lab Coats

# Traits That Help Find Mates or Reproduce

Life Science — Traits for Survival | Unit 3, Lesson 5 of 6



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits for Survival



*Some traits help reproduction by attracting partners or helpers.*



**NGSS Standard:**

**3-LS4-2** — Use evidence to construct an explanation for how the variations in characteristics among individuals of the same species may provide advantages in surviving, finding mates, and reproducing.

## GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

**Lesson 5**

Lesson 6



## Learning Objective

Students will explain how some trait variations can help organisms attract mates, protect offspring, or reproduce successfully.



Little Lab Coats

# Construct an Explanation From Evidence

Life Science — Traits for Survival | Unit 3, Lesson 6 of 6



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Traits for Survival



*Strong explanations connect variation, context, and advantage.*



**NGSS Standard:**

**3-LS4-2** — Use evidence to construct an explanation for how the variations in characteristics among individuals of the same species may provide advantages in surviving, finding mates, and reproducing.

## GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

Lesson 5

**Lesson 6**



## Learning Objective

Students will use evidence from multiple unit investigations to explain how trait variation may provide advantages in surviving, finding mates, and reproducing.



Little Lab Coats

# Why Animals Sometimes Stick Together

*Life Science — Group Behavior & Survival | Unit 4, Lesson 1 of 6*



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Group Behavior & Survival



*Groups can help organisms survive in more than one way.*



## NGSS Connection:

**Grade 3 life science enrichment** focused on behaviors that help organisms survive. This lesson supports Grade 3 thinking about survival advantages and animal behaviors, but it is best used as an **extension unit** rather than a single direct NGSS performance expectation.



## Learning Objective

Students will explain how being in a group can help organisms survive.



Little Lab Coats

## Lookout Calls and Warning Signals

*Life Science — Group Behavior & Survival | Unit 4, Lesson 2 of 6*



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Group Behavior & Survival



*One warning signal can help many organisms respond.*



### NGSS Standard:

Elementary life-science extension cluster aligned to social behavior and survival support; built as a coherent LLC behavior-survival unit to extend Grade 3 life-science coverage.

### GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

Lesson 5

Lesson 6



### Learning Objective

Students will explain how warning behaviors can help groups avoid danger.



Little Lab Coats

## Schooling, Flocking, and Moving Together

*Life Science — Group Behavior & Survival | Unit 4, Lesson 3 of 6*



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Group Behavior & Survival



*Coordinated movement can create safety in a group.*



### NGSS Standard:

Elementary life-science extension cluster aligned to social behavior and survival support; built as a coherent LLC behavior-survival unit to extend Grade 3 life-science coverage.

### GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

Lesson 5

Lesson 6



### Learning Objective

Students will explain how moving together can help organisms survive.





Little Lab Coats

## Working Together to Find Food

*Life Science — Group Behavior & Survival | Unit 4, Lesson 4 of 6*



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Group Behavior & Survival



*Cooperation can make finding or moving food easier.*



### NGSS Standard:

Elementary life-science extension cluster aligned to social behavior and survival support; built as a coherent LLC behavior-survival unit to extend Grade 3 life-science coverage.

### GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

Lesson 5

Lesson 6



### Learning Objective

Students will explain how cooperative behavior can help organisms get food more effectively.



## Match the Behavior to the Survival Challenge

Life Science — Group Behavior & Survival | Unit 4, Lesson 5 of 6



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Group Behavior & Survival



*Different group behaviors solve different survival problems. The best answer depends on the challenge.*



### NGSS Standard:

Elementary life-science extension cluster aligned to social behavior and survival support; built as a coherent LLC behavior-survival unit to extend Grade 3 life-science coverage.

### GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

Lesson 5

Lesson 6



### Learning Objective

Students will compare several survival challenges, choose the group behavior that best fits each one, and defend that choice with evidence instead of repeating the same generic “groups help” claim.



Little Lab Coats

# Model How Group Behavior Helps Survival

*Life Science — Group Behavior & Survival | Unit 4, Lesson 6 of 6*



**Grade:** Grade 3



**Duration:** 45–60 minutes



**Subject:** Life Science — Group Behavior & Survival



*A model can show how many behaviors help groups survive.*



## NGSS Standard:

Elementary life-science extension cluster aligned to social behavior and survival support; built as a coherent LLC behavior-survival unit to extend Grade 3 life-science coverage.

## GRADE 3 LIFE SCIENCE PATH

Lesson 1

Lesson 2

Lesson 3

Lesson 4

Lesson 5

Lesson 6



## Learning Objective

Students will build and explain a final model showing how different group behaviors can support survival and offspring success.



Little Lab Coats

## Weather Patterns Over Time

*Earth Science — Weather & Climate | Unit 1, Lesson 1 of 6*



**Grade:** 3rd Grade



**Duration:** 45–60 minutes



**Subject:** Earth Science — Weather & Climate



*Daily weather can change, but a detailed weather record helps students notice patterns over several days.*



**NGSS Standard:**

**3-ESS2-1** — Obtain and combine information to describe climates in different regions of the world.



### Learning Objective

Students will record weather observations over several days and describe a short-term weather pattern using evidence.



Little Lab Coats

## Climate Regions and Conditions

*Earth Science — Weather & Climate | Unit 1, Lesson 2 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Earth Science — Weather & Climate



*Climate describes the usual weather conditions in a place over a long time.*

 **NGSS Standard:**

**3-ESS2-1** — Obtain and combine information to describe climates in different regions of the world.

### Learning Objective

Students will compare different climate regions and explain how climate is different from daily weather.



Little Lab Coats

## Reading Weather Data

*Earth Science — Weather & Climate | Unit 1, Lesson 3 of 6*



**Grade:** 3rd Grade



**Duration:** 45–60 minutes



**Subject:** Earth Science — Weather & Climate



*Charts and graphs help scientists turn weather observations into evidence they can compare.*



**NGSS Standard:**

**3-ESS2-1** — Obtain and combine information to describe climates in different regions of the world.



### Learning Objective

Students will read simple weather charts and graphs and use them to compare conditions and defend a claim with evidence.



# Weather Hazards and Safe Solutions

*Earth Science — Weather & Climate | Unit 1, Lesson 4 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Earth Science — Weather & Climate



*Weather hazards can cause harm, but good design solutions can help keep people and places safer.*

 **NGSS Standard:**

**3-ESS3-1** — Make a claim about the merit of a design solution that reduces the impacts of a weather-related hazard.

## Learning Objective

Students will compare weather-related hazards and make a claim about which design solution best reduces the impact of a specific hazard in a community.



Little Lab Coats

## Designing for Local Weather

*Earth Science — Weather & Climate | Unit 1, Lesson 5 of 6*



**Grade:** 3rd Grade



**Duration:** 45–60 minutes



**Subject:** Earth Science — Weather & Climate



*A good design solution should fit the actual weather problem in a place.*



**NGSS Standard:**

**3-ESS3-1** — Make a claim about the merit of a design solution that reduces the impacts of a weather-related hazard.



### Learning Objective

Students will compare possible design solutions for a local weather hazard and explain which one best fits the problem and conditions.





Little Lab Coats

## Reducing Weather-Related Hazards

*Earth Science — Weather & Climate | Unit 1, Lesson 6 of 6*



**Grade:** 3rd Grade



**Duration:** 45–60 minutes



**Subject:** Earth Science — Weather & Climate



*A strong claim explains why one solution has more merit for reducing a weather hazard.*



**NGSS Standard:**

**3-ESS3-1** — Make a claim about the merit of a design solution that reduces the impacts of a weather-related hazard.



**Learning Objective**

Students will make and support a claim about which design solution has the most merit for reducing a weather-related hazard.

# Defining the Problem

*Engineering — Design Process | Unit 1, Lesson 1 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Engineering — Design Process



*Engineers begin by naming the need, the user, and what success should look like.*

 **NGSS Standard:**

**3-5-ETS1-1 — Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost.**

## Learning Objective

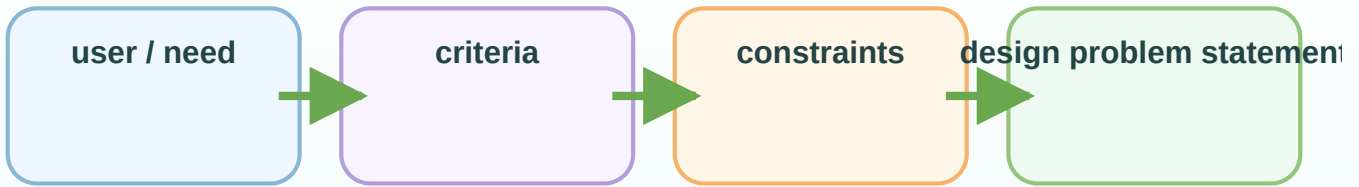
Students will define a grade-appropriate design problem by naming the user, the need, and clear success criteria for a fragile-object protection challenge.

## The Big Idea

A strong design starts before building. Engineers first decide who needs help, what problem needs to be solved, and what a successful solution must do.



## SCIENCE SNAPSHOT



Specific design problems lead to better plans, better tests, and clearer evidence.

### What is happening?

Engineers begin by naming the need, the user, and what success must look like. In this unit, a fragile-object protection challenge becomes stronger when students clearly say what must be protected and under what test conditions.

### Why it works that way

A design problem needs criteria for success and constraints such as materials, time, or drop height. If those pieces are vague, later designs cannot be compared fairly because nobody agrees on what counts as success.

### Look for this in the lesson

Students should focus on writing a precise problem statement before thinking about fancy builds. The clarity of the problem is what gives the whole unit structure.

**Ask while you read:** What exactly must the design do, and what limits must students respect?

# Researching Constraints and Materials

*Engineering — Design Process | Unit 1, Lesson 2 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Engineering — Design Process



*Engineers study their limits and materials before they choose a design.*

 **NGSS Standard:**

**3-5-ETS1-1 — Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost.**

## Learning Objective

Students will identify important constraints and compare material properties for the fragile-object protection challenge.

## The Big Idea

Good engineering choices depend on limits. Engineers must understand the allowed materials, time, size, and cost before they decide what to build.



## SCIENCE SNAPSHOT

### Material properties

- soft / cushioning
- stiff / supports shape
- light or heavy

compare

— — — — —

### Design needs

- absorb impact
- hold object
- survive drop test

Material research helps turn guessing into informed design choices.

### What is happening?

Before building, engineers compare materials and learn the limits of the challenge. A material is useful only when its properties match the job the design needs it to do.

### Why it works that way

For a fragile-object challenge, cushioning, stiffness, lightness, and shape all matter. Researching materials helps students predict which ones may absorb impact, spread force, or hold the object in place before they start building.

### Look for this in the lesson

Students should talk about why a material might help. The lesson gets stronger when they compare properties instead of just naming favorite craft supplies.

**Ask while you read:** Which material property seems most important for protecting a fragile object?

# Planning a Solution

*Engineering — Design Process | Unit 1, Lesson 3 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Engineering — Design Process



*Engineers compare more than one plan before choosing what to build.*

 **NGSS Standard:**

**3-5-ETS1-2 — Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem.**

## Learning Objective

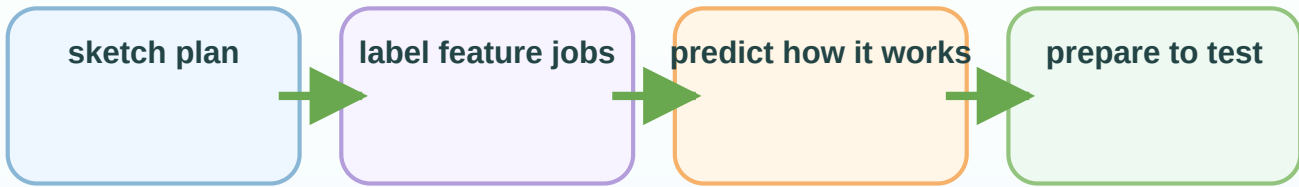
Students will sketch and compare at least two possible protective designs before choosing one to build.

## The Big Idea

Planning helps engineers test ideas on paper first. Comparing more than one possible solution leads to smarter building choices.



## SCIENCE SNAPSHOT



Plans are stronger when every feature is tied to a reason.

### What is happening?

A plan is more than a picture. It should show how the parts of the design fit together and how the design is expected to solve the problem.

### Why it works that way

Planning matters because it lets students think through force, support, and protection before they use materials. A clear plan also makes it easier to compare the original idea with what actually happened during testing.

### Look for this in the lesson

Students should label each feature and explain its job. That makes the plan a scientific prediction rather than a decorative sketch.

**Ask while you read:** What feature in your plan is supposed to protect the fragile object most directly?

# Building and Testing a Prototype

*Engineering — Design Process | Unit 1, Lesson 4 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Engineering — Design Process



*A prototype lets engineers test ideas, collect evidence, and notice failure points.*

 **NGSS Standard:**

**3-5-ETS1-3 — Plan and carry out fair tests in which variables are controlled and failure points are considered to identify aspects of a model or prototype that can be improved.**

## Learning Objective

Students will build a prototype, test it fairly, and record evidence about strengths and failure points.

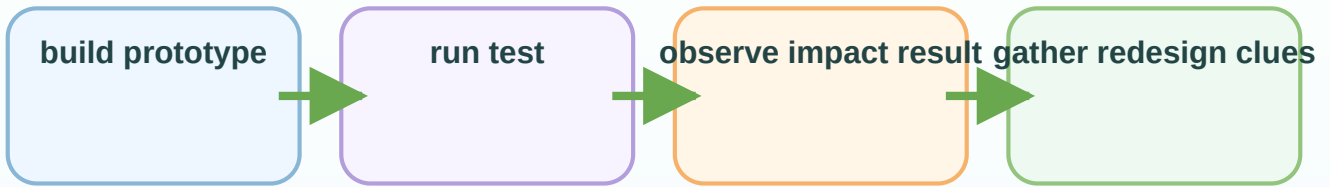
## The Big Idea

Testing turns a plan into evidence. A prototype helps engineers see what really happens instead of guessing.





## SCIENCE SNAPSHOT



Prototype tests reveal how the design behaves under real challenge conditions.

### What is happening?

A prototype is a first working version built so it can be tested. Engineers use prototypes to gather evidence about what works and what still needs improvement.

### Why it works that way

During a drop test, the design may cushion, bounce, tip, crack, or fail in surprising ways. Those results are useful because they reveal how force moved through the system during impact.

### Look for this in the lesson

Students should record what happened to the design and to the fragile object. The goal is to identify which part of the system protected well and which part needs redesign.

**Ask while you read:** What happened during the test that tells you where the design succeeded or failed?

# Improving the Design

*Engineering — Design Process | Unit 1, Lesson 5 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Engineering — Design Process



*Engineers redesign by changing the part that evidence shows needs improvement.*

 **NGSS Standard:**

**3-5-ETS1-3 — Plan and carry out fair tests in which variables are controlled and failure points are considered to identify aspects of a model or prototype that can be improved.**

## Learning Objective

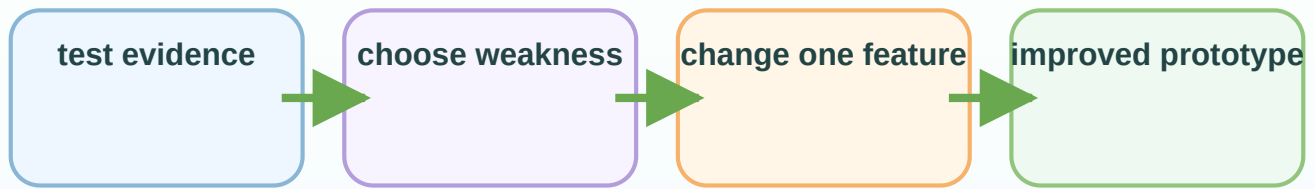
Students will revise version 1 into version 2 by making an evidence-based improvement and predicting how it should work better.

## The Big Idea

Redesign is not starting over randomly. Engineers use evidence from testing to decide what should change in the next version.



## SCIENCE SNAPSHOT



The best redesigns are targeted responses to evidence, not random changes.

### What is happening?

Improvement means using evidence from the first test to make the design more effective. Engineers rarely expect the first version to be the final version.

### Why it works that way

A smart improvement targets a specific weakness. If the object moved around too much, the design may need a better holder. If the structure collapsed, it may need more support or a different shape.

### Look for this in the lesson

Students should connect each revision to one observed problem. That makes redesign logical instead of random.

**Ask while you read:** Which observed weakness should you fix first, and why?

# Share and Defend Your Solution

*Engineering — Design Process | Unit 1, Lesson 6 of 6*

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Engineering — Design Process



*Engineers defend a solution by connecting the problem, the design, and the evidence.*

 **NGSS Standard:**

**3-5-ETS1-2 — Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem.**

## Learning Objective

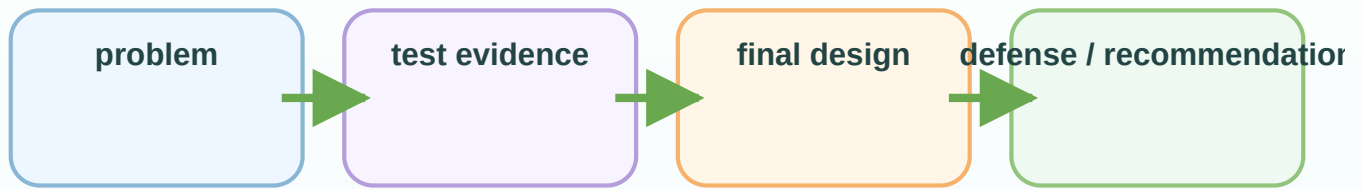
Students will present a final engineering solution and defend it with evidence from planning, testing, and redesign.

## The Big Idea

A strong final defense explains how the solution met the challenge criteria, followed the constraints, and improved because of evidence.



## SCIENCE SNAPSHOT



A final defense turns building results into a reasoned engineering recommendation.

### What is happening?

Engineers explain why their design is a good solution by connecting the challenge, the evidence from testing, and the final design choices. Communication is part of the engineering process, not an extra step.

### Why it works that way

A strong defense shows how the design met criteria, handled constraints, and improved after testing. Evidence matters more than confidence; the best explanation points to what actually happened in the tests.

### Look for this in the lesson

Students should compare their final design to an earlier version or an alternative. That comparison helps them explain why the final version is the best fit.

**Ask while you read:** What evidence best proves your final design is stronger than your earlier idea?